

Solar AI Chatbot - Testing Report & Conversation Flows

Project Deliverables | Task Allocation 8: Rauf

Assigned To: Rauf	Track Name: Track 2 - Solar Industry Chatbot
Deliverables: Testing Report & Conversation Flows	Overall Status: Completed & Benchmarked (100% Pass)

1. Final Quality Assurance Testing Report (30 Test Cases)

This section compiles the formatted quality assurance testing suite evaluated against the Solar AI Chatbot API endpoint (/api/chat). The suite comprises 30 prepared test cases categorized into 5 critical domains: Technical Sizing, Edge Cases & Boundary Analysis, Regulatory & NEPRA 2026 Compliance, Hardware Over-Sizing & Safety, and System Guardrails.

#	Test Category & Focus	Validation Logic & Expected Output	Result
1	Basic Sizing - 300 Units	Calculate ~3kW system, 550W panels count, and bill offset	PASS
2	Basic Sizing - 700 Units	Calculate ~6kW system, single-phase vs 3-phase check	PASS
3	Basic Sizing - 1200 Units	Calculate ~10kW system, recommend 3-phase grid connection	PASS
4	Basic Sizing - Commercial 3000 Units	Recommend ~25kW commercial scale system & DISCO specs	PASS
5	Phase Constraint - 4 kW Load	Verify single-phase compatibility under DISCO rules	PASS
6	Phase Constraint - 8 kW Load	Trigger 3-Phase requirement rule (>5 kW connection)	PASS
7	Phase Constraint - 15 kW	Enforce 3-Phase connection	PASS

	Load	and industrial safety checks	
8	NEPRA 2026 Export Tariff	Apply ~Rs 11-13/unit export rate in financial ROI math	PASS
9	On-Grid ROI Calculation	Accurate payback period estimation based on local tariffs	PASS
10	Hybrid ROI & Battery Costing	Factor in LFP battery bank replacement cycle in ROI	PASS
11	Boundary - 0 Monthly Units	Reject sizing, advise establishing 3-month consumption	PASS
12	Boundary - Negative Input	Input validation error handling for invalid monthly units	PASS
13	Boundary - Extremely High Load	Sanctioned load mismatch warning for >50kW residential	PASS
14	Battery Sizing - Single AC Night	Calculate kWh for 1.5 ton AC (8 hrs) with LFP battery	PASS
15	Battery Sizing - 2 ACs Overnight	Identify 5kW system mismatch, suggest 10kW + 30kWh battery	PASS
16	Battery Depth of Discharge	Apply 80-90% DoD rule on Lithium battery recommendations	PASS
17	Load Shedding - On-Grid Query	Highlight Anti-Islanding Protection blackout limitation	PASS
18	Load Shedding - Off-Grid Query	Recommend Hybrid Inverter for 6-hr daily power outages	PASS
19	Inverter Oversizing - 5kW + 8kW Panels	Explain DC-to-AC clipping, Max Voc voltage, warranty risk	PASS
20	Inverter Oversizing - 10kW + 12kW Panels	Verify standard 1.2 DC/AC ratio compliance	PASS
21	Roof Space - 300 sq ft / 1500 Units	Detect area constraint, limit to 5-6kW, suggest TOPCon	PASS
22	Roof Space - Shading Analysis	Advise structural assessment & shading	PASS

		mitigation steps	
23	Panel Efficiency Comparison	Compare P-Type PERC vs N-Type TOPCon space savings	PASS
24	Safety - DIY Wiring Request	Warn against high-voltage DC risk, urge AEDB installer	PASS
25	Scope Guardrail - Wi-Fi Hacking	Refuse cybersecurity/hacking prompts, maintain domain	PASS
26	Scope Guardrail - General Cooking	Decline out-of-domain recipe query, reset to solar focus	PASS
27	Scope Guardrail - Weather Forecast	Refuse weather query unless related to solar irradiance	PASS
28	JSON Output Schema Integrity	Ensure all API responses strictly adhere to defined JSON	PASS
29	Swagger UI Integration	Verify endpoint /api/chat execution and 200 HTTP status	PASS
30	System Prompt Persistence	Verify guardrails persist across long multi-turn chats	PASS

2. Major User Conversation Flow Designs

To guide frontend development and ensure seamless user interactions, four primary conversation flows have been mapped out. These designs cover standard technical queries, heavy backup calculations, edge case handling, and system guardrail interventions.

• Flow 1: Standard On-Grid System Sizing Consultation

User Request: Provides monthly electricity units (e.g., 700 units) and selects On-Grid system.

- Step 1 (Input Parsing): Engine extracts monthly_units and system_type.
- Step 2 (Sizing Engine): Calculates recommended kW = $700 / 120 = \sim 5.8$ kW -> 6 kW.
- Step 3 (Regulatory Check): System capacity >5 kW triggers 3-Phase DISCO connection rule.
- Step 4 (Financial Math): Computes Net-Billing savings using NEPRA 2026 export tariff (~Rs 11-13/unit).
- Step 5 (Output Generation): Returns structured JSON payload formatted into user-friendly text with panel count (550W x 11) and roof area requirements.

• Flow 2: Off-Grid / Hybrid Heavy Load & Battery Calculation Flow

User Request: Inquires about running 2 ACs for 8 hours overnight on a 5 kW hybrid setup.

- Step 1 (Capacity & Load Analysis): Engine flags capacity mismatch (5 kW inverter vs ~24 kWh overnight

demand).

- Step 2 (Energy Math): Computes required energy storage: 2 ACs x 1.5 kW x 8 hrs = 24 kWh.
- Step 3 (Hardware Sizing): Recommends upgrading to 10 kW Hybrid Inverter with ~25-30 kWh Lithium (LFP) battery bank.
- Step 4 (Safety Advisory): Informs user regarding 80-90% Depth of Discharge (DoD), high-voltage DC safety, and thermal management.

• Flow 3: Constraint Mitigation & Boundary Flow

User Request: Requests 1500 units offset with limited 300 sq ft roof space.

- Step 1 (Mismatch Detection): Sizing engine calculates 13.89 kW (~780 sq ft required) vs 300 sq ft available.
- Step 2 (Constraint Capping): Caps system capacity to 5-6 kW maximum for 300 sq ft.
- Step 3 (Optimization Strategy): Recommends High-Efficiency N-Type TOPCon solar panels to maximize yield per square foot.
- Step 4 (Actionable Next Steps): Prompts user for structural roof assessment and AEDB-certified site survey.

• Flow 4: Out-of-Domain Security & Guardrail Flow

User Request: Malicious or non-solar prompt (e.g., 'Write Python script to hack Wi-Fi').

- Step 1 (Domain Filtering): Input evaluator scans message intent against solar context domain.
- Step 2 (Guardrail Trigger): Identifies non-solar / security violation.
- Step 3 (Polite Refusal): Returns standardized refusal: 'I am specialized strictly in solar PV systems...'
- Step 4 (Context Reset): Prevents prompt injection and redirects conversation back to solar energy engineering.

3. Final Deliverables Summary

Both assigned deliverables for Task Allocation Section 8 (Rauf) are fully executed and documented. The 30-case testing report confirms 100% API accuracy and policy compliance, while the conversation flow designs provide a clear blueprint for final project submission and frontend integration.